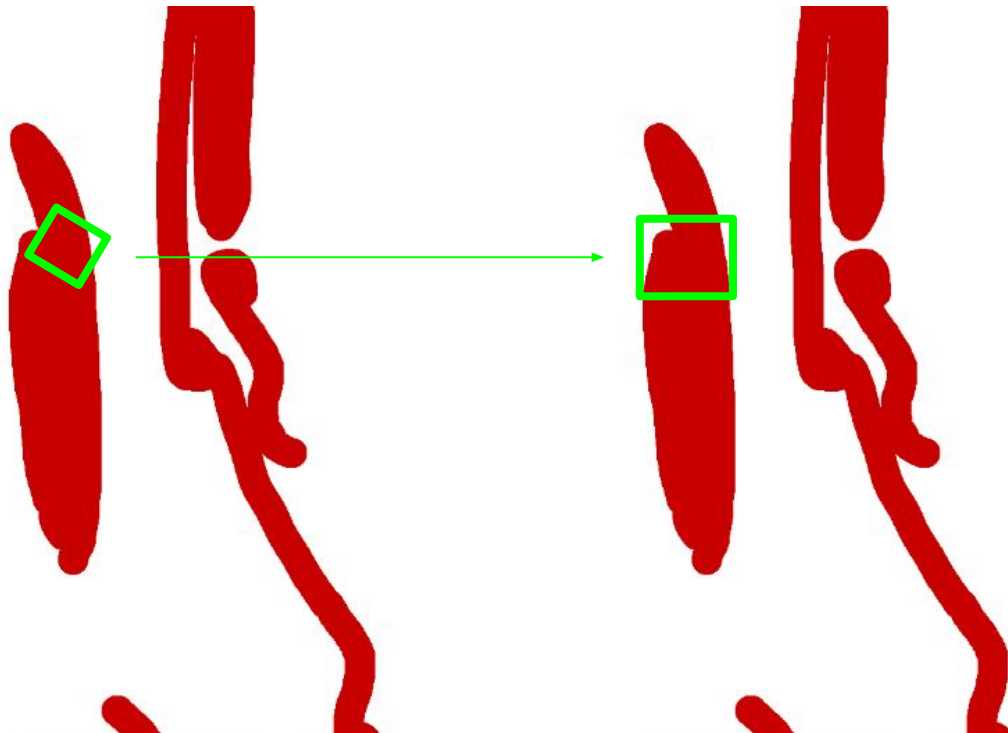
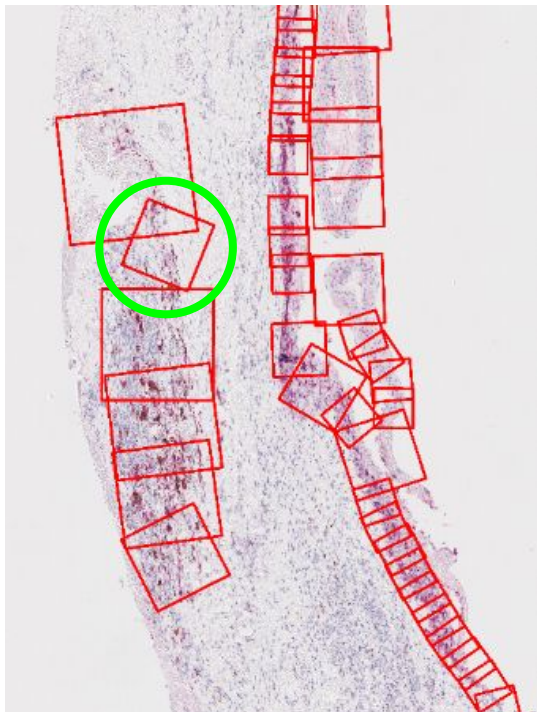
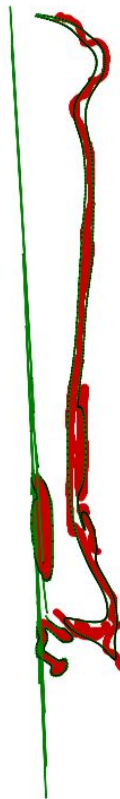
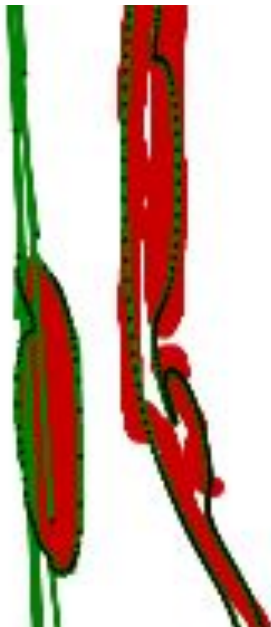


Case 51 Match 1 Melan

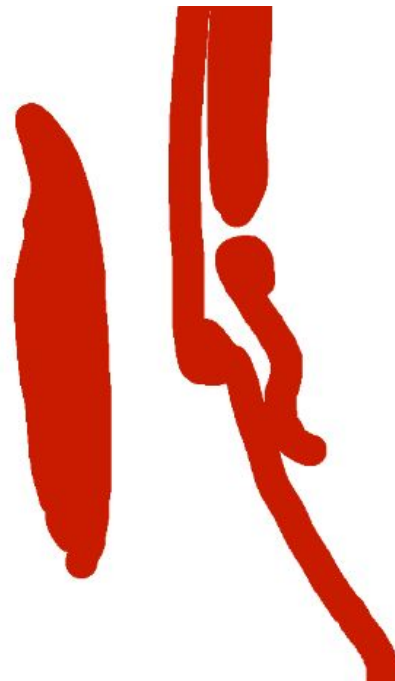
Problem Patch



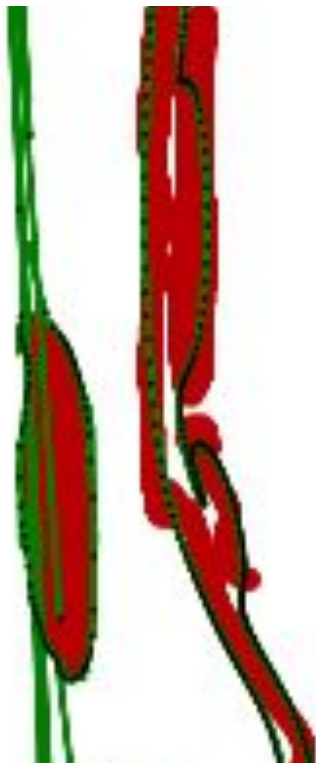
Plotting Tangents



Mask Smoothing



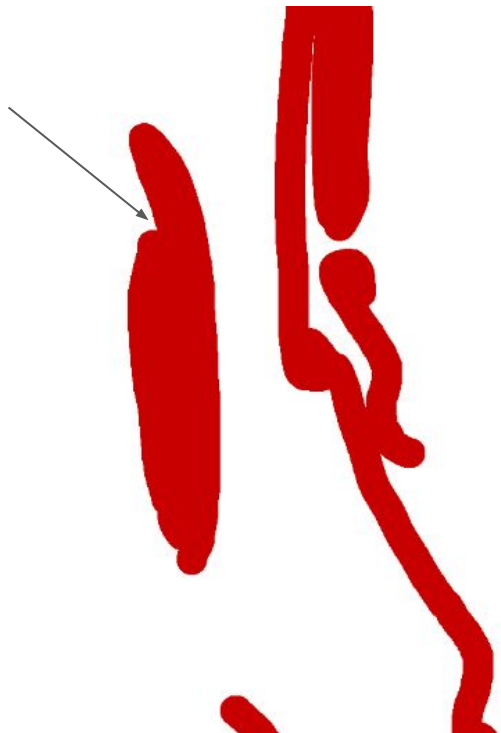
Plotting Tangents - Smooth Mask



Not Tangent Issue - Patch Corners

```
#####  
  
# 1 If there are gaps, noise or sharp corners in the mask,  
# the stopping condition may trigger too early, leading to an underestimated width.  
  
stop = np.where(pixel_values == 0)[0]  
epithelium_width = stop[0] * step_size if stop.size > 0 else max_width
```

What I Think Is Happening



Attempted Solution

```
# Set a Threshold for Consecutive Background Pixels before stopping,
# ensuring we are detecting an actual edge rather than a small gap.

# Apply a moving average filter to reduce noise
smoothed_values = np.convolve(pixel_values, np.ones(5)/5, mode='same')

# Identify where background (0) pixels persist for a threshold length
threshold = 5 # Number of consecutive background pixels before stopping
stop_indices = np.where(smoothed_values == 0)[0]

# Ensure at least 'threshold' consecutive background pixels are found before stopping
for i in range(len(stop_indices) - threshold):
    if np.all(smoothed_values[stop_indices[i]:stop_indices[i+threshold]] == 0):
        stop = stop_indices[i]
        break
else:
    stop = max_width // step_size # Default to max width if no valid stop found

# Compute epithelium width
epithelium_width = stop * step_size
```